

Alli-Trade

SW Engineering CSC648-848 Spring 2026

Team 17

Milestone 1

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History Table

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1. Executive Summary

San Francisco State University students often need affordable ways to buy and sell everyday items such as textbooks, furniture, electronics, and services. While platforms like Facebook Marketplace or Craigslist exist, they are not tailored to the SFSU community and often involve risks such as scams, unreliable sellers, or meeting strangers outside the campus network. Students also frequently need services such as tutoring, moving help, or small freelance work that are difficult to advertise safely within the university community. Alli Trade is designed to address this gap by creating a dedicated marketplace exclusively for SFSU students, providing a safer, more convenient way to exchange goods and services within a trusted campus network.

Alli Trade is a web application that allows SFSU students to post items for sale, browse listings, offer services, and communicate directly with other students. The platform functions similarly to a simplified campus version of Facebook Marketplace but is built specifically for the needs of university students. Users can create listings for items they wish to sell, place bids on listings, offer services such as tutoring or design work, and communicate through an integrated messaging system. Importantly, the platform does not process payments directly; instead, it facilitates communication between buyers and sellers so they can arrange transactions independently. This approach keeps the platform simple while giving students a convenient way to connect with others on campus.

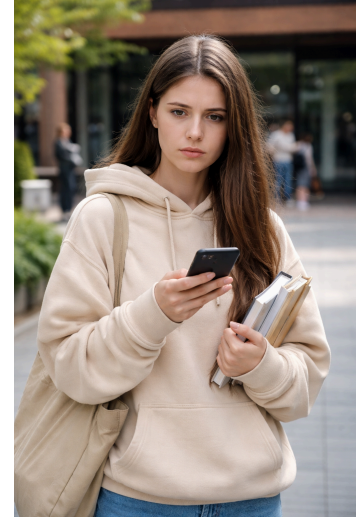
What makes Alli Trade unique is its focus on the SFSU student community. Access is limited to SFSU students, helping create a trusted environment where users know they are interacting with members of their own campus community. The platform is designed specifically for student needs: affordable second-hand goods, service exchanges, and local meetups that can occur safely on campus. By keeping the marketplace focused on SFSU, we can offer a convenient, campus-focused exchange medium to students.

Alli Trade is being developed by Team 17 at San Francisco State University as part of a collaborative software engineering project. Our team combines skills in software development, web design, and system deployment to build a reliable and scalable platform using modern web technologies. By building this platform ourselves, we gain real-world experience in designing, deploying, and maintaining a full-stack web application while creating a tool that can directly benefit the SFSU community. Our goal is to deliver a useful, student-focused marketplace that improves how students connect, trade, and support each other on campus.

2. Personae

2.1 Audrey the aloof student (Inexperienced/high friction buyer)

Audrey is not interested in understanding how technology works and spends as little time on devices as possible. Convenience is very important to her. She doesn't trust online payments and is drawn to using the application due to its promise that no financial information must be provided to use the marketplace. One of her primary concerns is physical safety when exchanging items with strangers: she has never done it before. She is willing to use the platform as it only includes users from a vetted community and promotes a monitored location where she is able to meet to complete a transaction.

**Characteristics:**

- New to platform
- High friction buyer
- Cautious/untrusting of strangers

Goals:

- Complete exchange transaction
- Feel safe about transacting
- Protect personal information

Skills:

- Email
- Desktop user

Pain Points:

- Difficulty with mobile phones
- Provides limited personal data
- Not communicative, low engagement

2.2 Sally the efficient student (ideal user)

Sally is a junior student at SFSU majoring in computer science while living on campus. One of their issues is that they have too much stuff. They want to move out and need to get rid of a few items. This person is productive and efficient, they use our app for the sole purpose of selling their items instead of casually browsing. Sally is also tech oriented and a fast learner, they easily pick up the software, create an account and start selling stuff immediately. As soon as they start posting their stuff for sale she now has to handle multiple sell requests and even has trouble managing all of these sell requests.



Characteristics:

- Very productive, efficient and goal oriented
- Less likely to be distracted by browsing and shopping

Skills:

- Tech oriented
- Fast learner

Pain points:

- Managing too many sell requests.
- Has no reliable transportation to obtain items

2.3 Maya Chen the Peer Tutor (service seller)

Maya Chen is a 20-year-old junior at San Francisco State University majoring in Psychology. She does well in her statistics and research methods classes and wants to earn some extra money while helping other students succeed. To reach students who might need help studying, she uses the campus marketplace app to advertise affordable tutoring sessions to others in the university community.



Characteristics:

- Undergraduate student
- Lives off-campus in San Francisco
- Uses smartphone / laptop regularly

Goals:

- Offer tutoring services
- Coordinate easily with students
- Reach potential clients
- Earn extra income

Skills:

- Comfortable with common apps and marketplaces
- Able to create listings and respond to messages
- Confident explaining academic material

Pain Points:

- Difficulty reaching students who need help,
- Tutoring requests scattered across multiple platforms
- Unclear demand for subject areas

2.4 Lucas The Budget Conscious Buyer

Lucas is a budget-conscious buyer; he mainly uses the marketplace to save money. Lucas usually looks for low-priced second-hand items that are still in good condition. He hopes to be able to quickly find what they need without wasting time, getting scammed, or traveling far off campus.

Characteristics:

- Undergraduate student
- Lives in a dorm on campus
- Shops for practical, rather than luxury

Goals:

- Finds affordable items (Often compares multiple listings)
- Buys locally
- Avoid getting scammed

Skills:

- Familiar with online shopping behavior
- Can search, filter, and compare listings

Pain Points:

- Unclear or misleading listings
- Listings with items that have already been sold



3. High-Level Use Cases

3.1 – Selling Tutoring Services

Actor: Maya (Junior, Psychology)

Maya Chen, a junior student at San Francisco State University, wants to earn extra money by tutoring classmates who struggle with statistics and psychology research methods. As a typical student with everyday technology skills, Maya uses the campus marketplace to post a tutoring service so other students can find her. Students browsing the platform for academic help see her listing and contact her to ask questions about the subjects she tutors. Maya communicates with interested students through the platform and arranges a time to meet for tutoring, allowing both Maya and her classmates to connect easily within the campus community for academic support.

3.2 – Completing Transactions Safely

Actor: Audrey (Junior, Musicology)

Audrey wants to sell some textbooks from past classes that she doesn't think are worth holding onto. She wants to be able to include all the textbooks in a single post, and then communicate with the potential buyers through an email set up specifically for this exchange. She wants to upload photos of the textbooks from her phone, but would prefer to manage the transactions from her computer. All communication will need to be done through email, because she will only access the service to add and remove the post. The platform also provides her an exchange location that she feels is safe. She has never been there before and needs help navigating to her destination. At least one of her close friends will know where and when the transaction will take place. One of her primary concerns is physical safety when exchanging items with strangers: she has never done it before. She is willing to use the platform as it only includes users from a vetted community and promotes a monitored location where she is able to meet to complete a transaction.

3.3 – Facilitating Carpool and Rideshare

Actor: Marcus (Senior, Engineering)

Marcus lives in the East Bay and drives to the SFSU campus three days a week. To offset the high cost of gas and parking, he posts a rideshare service listing to the marketplace. Other students, like Maya, who have classes on the same days, find his post while browsing the transportation category. Maya initiates a conversation through the platform to verify Marcus's schedule and departure location. They use the built-in messaging to agree on a weekly contribution and a specific pick-up spot near the Daly City BART station. This allows Marcus to reduce his commuting expenses while providing Maya with a reliable, student-only transit alternative.

3.4 – Transitioning On-Campus Housing Supplies

Actor: Leo (Freshman, International Student)

Leo is moving into the Manzanita Square dorms and needs bulky essentials like a mini-fridge and a desk lamp, but he doesn't have a car to visit big-box retailers. He searches the marketplace specifically for "On-Campus Pickup" filters. He connects with Sally, who is moving out of the same building and looking to rehome her appliances quickly. They use the platform's real-time texting to coordinate a meetup right in the dorm lobby. Because the platform verifies their SFSU student status, Leo feels confident that the items are coming from a peer within his immediate living community. He meets Sally, inspects the fridge, and completes the move-in process without ever having to leave the university grounds.

4. Data Item Glossary

Term	Description
Unregistered User	A user that can browse the marketplace and view listings but must register and login to access other app features
Registered User	Someone accessing the marketplace that can also chat with other registered users , modify profile data, and create (and remove) listings
User	A term that encompasses both registered users and unregistered users .
Account	The registered users profile
Profile	The personal account the registered user makes when signing up for the marketplace for the first time. Contains the user's name, with optional visibility of the registered user's email, photo, major and graduation year or other active university role, and active listings . The registered users can add or delete posts from this page.
Marketplace	Term used to refer to the webpage of active listings up for sale
Register	The act of creating an account to have full access to chat functions
Buyer	Registered User interested in purchasing item/service
Seller	Registered User listing item or service .
Product	A physical item put up for sale by a seller
Service	A service or intangible item up for sale by a seller
Item	A generalized term to easily refer to a product or service
Listing	Refers to active items in the marketplace
Price	The amount of money a seller is requesting for an item
Post	The act of completing an item's necessary parts to be visible listing and up for sale on the marketplace .
Listing Photo	Refers to the digital photo(s) of a listing .
Description	A short paragraph describing the item for sale, made by the seller when posting an item on the marketplace

Profile Data	Photo, major, graduation year
Admin	A registered user with heightened privileges on the app.
Chat	The in app function of messaging other registered users . Primarily used for buyer-seller communications
Message	The act of starting a chat with another registered user .
Chat History	The log of previous chats made between a registered user and other registered users .

5. Functional Requirements

1. User management
 - 1.1. **Registered Users** are only allowed to use an email with @sfsu.edu to gain access to the app.
2. Registered Users
 - 2.1. **Registered Users** shall create **profiles**, with their name, profile picture, major and graduation year or other university role
 - 2.2. **Registered Users** shall post **listings** for **items (products or services)** they want to sell
 - 2.3. **Registered Users** shall log in and log out of their **accounts**
 - 2.4. **Registered Users** shall update their **profile** information
3. Unregistered Users
 - 3.1. **Unregistered Users** shall be able to browse **listings** on the **marketplace** without creating an **account**.
 - 3.2. **Unregistered Users** shall be able to **search** for **listings**
 - 3.3. **Unregistered Users** shall be made to **register** before being able to message other **registered users**.
4. Admin management
 - 4.1. **Admins** shall be a **Registered User**
 - 4.2. **Admins** shall have admin access levels (Moderator, Admin, Owner)
 - 4.3. **Moderators** shall be able to flag accounts, suspend accounts, hide listings
 - 4.4. **Admins** shall be able to add/remove moderators and ban accounts
5. Listings
 - 5.1. **Registered Users** shall be able to create a **listing**
 - 5.2. **Registered Users** shall be able to update (edit or remove) their current **listings**
 - 5.3. **Registered Users** shall be able to label a **listing** as a **product** or **service**, with a **price**, **picture(s)**, and a **description** for the **item** as well as label the **item** with 1 or more **categories**
6. Searching
 - 6.1. **Users** shall be able to search for **products/services** using a search bar
 - 6.2. **Users** shall be able to use **categories** to browse specific **products/services**
 - 6.3. **Users** shall be able to browse all **listings** with a set amount of **listings** per page
 - 6.4. **Users** shall be able to change pages
 - 6.5. **Users** shall be able to sort **listings** by **price**, **category**, **product**, or **service**
7. Messaging
 - 7.1. **Buyers** shall be able to **chat** with **sellers** via in-app text messaging

- 7.2. **Registered Users** shall be able to create and delete **chats** with other **users**
- 7.3. **Registered Users** shall have a **chat history** with all the **chats** they've used

6. Non-functional Requirements

1. Application shall be developed, tested and deployed using tools and cloud servers approved by Class CTO and as agreed in M0
2. Application shall be optimized for standard desktop/laptop browsers e.g. must render correctly on the two latest versions of two major browsers
3. All our selected application functions shall be rendered well on mobile devices (no native app to be developed)
4. Posting sales information and messaging to sellers shall be limited only to SFSU students
5. Critical data shall be stored in the database on the team's deployment server.
6. No more than 50 concurrent users shall be accessing the application at any time
7. Privacy of users shall be protected
8. The language used shall be English (no localization needed)
9. Application shall be very easy to use and intuitive
10. Application shall follow established architecture patterns
11. Application code and its repository shall be easy to inspect and maintain
12. Google analytics shall be used
13. No e-mail clients or chat services shall be allowed. Interested users can only message to sellers via in-site messaging. One round of messaging (from user to seller) is enough for this application
14. Pay functionality, if any (e.g. paying for goods and services) shall not be implemented nor simulated in UI.
15. Site security: basic best practices shall be applied (as covered in the class) for main data items
16. Media formats shall be standard as used in the market today
17. Modern SE processes and tools shall be used as specified in the class, including collaborative and continuous SW development and GenAI tools
18. The application UI (WWW and mobile) shall prominently display the following exact text on all pages "SFSU Software Engineering Project CSC 648-848, Spring 2026. For Demonstration Only" at the top of the WWW page Nav bar. (Important to not confuse this with a real application). You must use this exact text without any editing.

19. Competitive Analysis

Features	Facebook Marketplace	Craigslist	Ebay	Our Future Product
Community Marketplace	+	+	-	++
Product Bids	+	-	++	-
Private Chat	+	+	+	+
Service Offers	+	++	+	+
Exchange Assistance	~	-	-	+
Verified Accounts	+	-	+	++

+ feature exists; ++ superior feature; - does not exist; ~ some variant exists

Competitive Analysis Summary:

Our product will be more specialized towards a campus marketplace rather than a general resale platform. Compared with Facebook Marketplace or Craigslist, our platform will focus on combining verified SFSU student accounts with features tailored to student needs, such as item bidding on items being sold, tutoring service offerings, an in-app/web chat from the client to the seller, and monitored exchange assistance. The platform will provide a search, category browsing, a way to sort by service or products, and in-app messaging between buyers and sellers. Unlike other broader marketplaces, this gives our product a competitive advantage in which we are offering a safer, more relevant, and more organized environment, specifically for a more specialized student marketplace.

20. High-Level System Architecture

Host Server: AWS EC2 T3 Small

Operating System: AWS Linux 2023

Web Server: NGINX 1.28.2, Gunicorn 25.1.0

Database: MySQL 8.4

Front-end: Django Templates, Tailwind, HTML, CSS

Server side: Python 3.13

Back-end: Django 6.0.2

Browser support: Google Chrome 146.0.7680.65 and Safari 26.3

Analytics: Google Analytics 4

APIs:

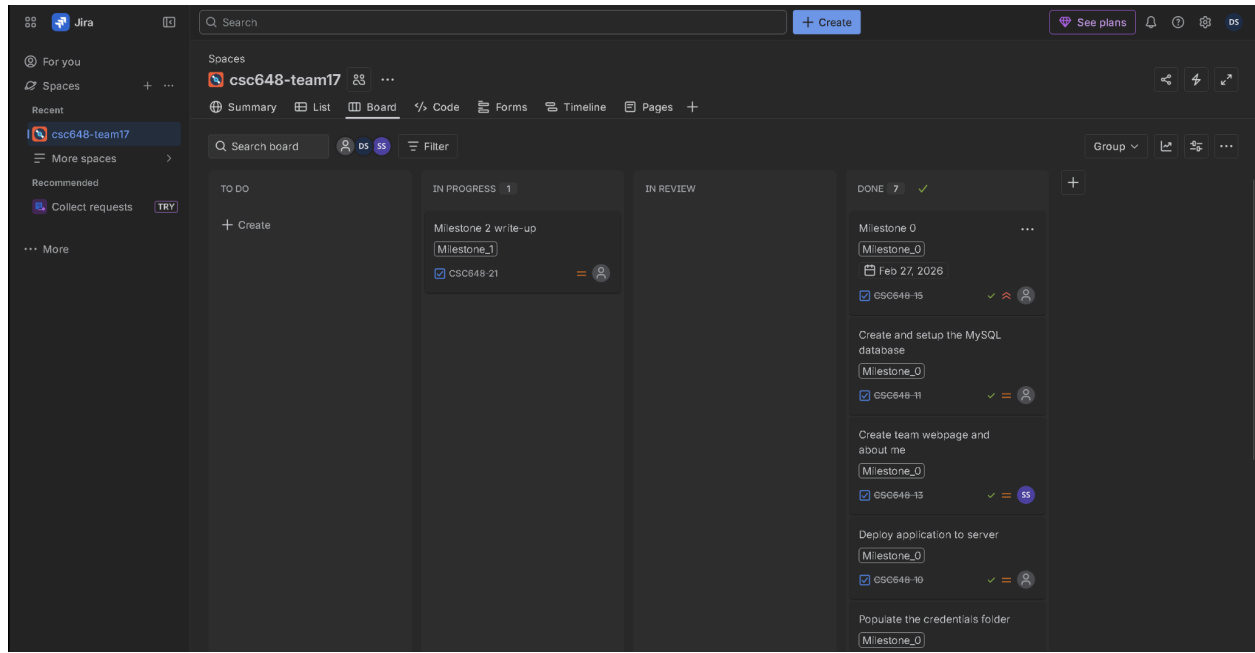
21. Sources/Gen AI

Version	Task / Used?	Rating	Explanation / Examples / Prompt / Comments
ChatGPT GPT-5.3 model	Persona / Used	High	- Utilized current free ChatGPT model to generate a persona based on a minimal prompt with some corrections - Generated content for Maya Chen - Content was mildly edited from final response from ChatGPT - Prompt: - I am developing an application for sfsu students to be able to buy and sell items to one another (this includes staff and faculty as well). I want to develop a use case. I want to focus this particular use case on someone aiming to find some cheap desks or other furniture for them to take to their home nearby campus. Please list their general characteristics, goals, skills, pain points related to the application. - Corrective Prompts: - The person should also have a name and a slight bio related to their goal on the app with a short story about them relative to this goal - Used to make the persona more personal and to create a short bio to “humanize” the persona it created - shorten the bio to about 3 sentences - Used to condense the multi-paragraph response generated - The original persona was general and had great structure and reasoning but was corrected to hone on the specifics of the project guidelines - The overall persona was changed from a product buyer to a service seller anyway, but the general framework was kept and remained helpful

GPT 5.3	Functional Requirements	Medium	- Utilized the current free ChatGPT model to generate some ideas for non-functional requirements - Prompt: - I am creating a buy and sell web application for SFSU students where they can list, purchase, and bid on both items and services. Create some function requirements for our web app. Drive this by use cases. *Provided use cases* Requirements have to satisfy/implement all use cases. Focus on WHAT and not HOW. Keep the user in mind. - Corrective Prompts: - List 15-20 requirements and organize them by category. - Make sure to use the word shall for all of them - The use cases were well-formatted and were appropriate based on our use cases. However, it did provide some that were out of scope for our level of depth. Was nice to have for brainstorming and reference, but ultimately made use cases on our own collaboratively.
Chat GPT 5.3	Personae Images	High	- Utilized the current free ChatGPT model to generate some images for each personae - Prompt: Using this personae *provided personae* create an image for the person - Corrective Prompt: - N/A - The images that were created were great and easy to add to our document. Happy with the results.

22. Task Management Tool Selection

The tool we selected: **Jira**



23. Team and Roles

Name	Email	Role
Subreet Singh	ssingh61@sfsu.edu	Team Lead
Lakshyaraj Bhati	lbhati@sfsu.edu	Team Member
Bikendra Shrestha	bshrestha1@sfsu.edu	Back-End Team Lead
Michal Krupa	mkrupa@sfsu.edu	Github Maintainer Front-End Team Member
Nicholas Blackson	nblackson@sfsu.edu	Document Maintainer Back-End Team Member
Daniel Smirnoff	dsmirnoff@sfsu.edu	Back-End Team Member
Jonathan Mai	jmai10@sfsu.edu	Front-End Team Member

24. Team Lead Checklist

So far all team members are fully engaged and attend team sessions when required

- *Done - **On Track** - Issue*

Team found a time slot to meet outside of the class

- ***Done** - On Track - Issue*

Team ready and able to use the chosen back and front-end frameworks and those who need to learn are working on learning and practicing

- ***Done** - On Track - Issue*

Team reviewed class slides on requirements and use-cases before drafting Milestone 1

- ***Done** - On Track - Issue*

Team reviewed non-functional requirements from “How to start...” document and developed Milestone 1 consistently

- ***Done** - On Track - Issue*

Team leads checked Milestone 1 document for quality, completeness, formatting and compliance with instructions before the submission

- ***Done** - On Track - Issue*

Team lead ensured that all team members read the final M1 and agree/understand it before submission

- ***Done** - On Track - Issue*

Team shared and discussed experience with GenAI tools among themselves

- ***Done** - On Track - Issue*

GitHub organized as discussed in class (e.g. master branch, development branch, folder for milestone documents etc.)

- ***Done** - On Track - Issue*

Team has chosen a task management tool and invited Course instructor to project via ajsouza@sfsu.edu email.

- ***Done** - On Track - Issue*